

Module A+

Chen, Pelger, Zhu (2024)

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Papers and materials

Chen, L., Pelger, M., & Zhu, J. (2024).
Deep Learning in Asset Pricing.
Management Science, 70(2), 714–750.

After Gu and the robustness block. 30 min, slides and discussion.

Authors' code: `github.com/LouisChen1992/Deep_Learning_in_Asset_Pricing`

Homework: `notebooks/04_chen_pelger_zhu_sketch.ipynb`

Paper card: Chen, Pelger & Zhu (2024)

	<i>Management Science</i> 70(2): 714–750
Question	Can a deep-learning stochastic discount factor (SDF), disciplined by no-arbitrage, price individual stocks better than pure return prediction?
Data	CRSP monthly, 1967–2016; 46 firm characteristics; 178 macro series.
Method	A GAN: a feed-forward net gives the SDF weights, an LSTM extracts a few macro states, and an adversarial net picks the test assets that are hardest to price.
How it is tested	Train 1967–86, validate 1987–91, test 1992–2016. Out-of-sample SDF Sharpe ratio, explained variation and pricing errors, against a forecasting benchmark.
Main result	Out-of-sample SDF Sharpe ratio of about 2.6 a year, almost twice the forecasting benchmark. It is still 1.73 when the 40% smallest stocks get zero weight.
Watch for	The loss targets average returns (first moments), not variance: the same distinction as R^2 vs GRS in our Fama–French notebook.
Code & data	Code on GitHub (LouisChen1992/Deep_Learning_in_Asset_Pricing); data need CRSP.

Why this paper after Gu

A pure prediction loss explains **variance** (a second moment).

The no-arbitrage condition targets **average returns** (a first moment).

It is the same distinction as R^2 vs GRS and mean $|\alpha|$ in notebook 01 .

Question: when does a finance constraint beat a pure ML loss?

Design in one slide

- CRSP monthly, 1967–2016 · 46 firm characteristics · 178 macro series
- Train 1967–1986 · validate 1987–1991 · **test 1992–2016**
- Three networks linked by the no-arbitrage condition:
 - feed-forward → SDF weights
 - LSTM → a few macro **states**
 - adversarial network → the **test assets** that are hardest to price
- Out-of-sample SDF Sharpe ratio $\approx$ **2.6** per year, almost twice the forecasting benchmark (FFN)

Gu vs Chen–Pelger–Zhu

	Gu–Kelly–Xiu (2020)	Chen–Pelger–Zhu (2024)
Object	predict $E[r]$	estimate the SDF
Objective	MSE / Huber; R_{OOS}^2	conditional moments, min–max (GMM-style)
Test assets	none in estimation; deciles for evaluation	chosen adversarially
Macro	8 predictors × characteristics	178 series → LSTM states
Size robustness	challenged by Avramov et al.	see next slide

Size and trading frictions (published version)

Fig. 17: set the SDF weights to zero below a cutoff (no re-estimation, so these are **lower bounds**)

Stocks removed	GAN annual Sharpe
none	≈ 2.6
40% smallest	1.73
40% widest bid–ask spread	2.07
40% least traded	1.87

Fn. 44: GKX-style long–short portfolios lose about half their Sharpe ratio when value-weighted: "a clear indication that the performance of these portfolios heavily depends on small stocks."

A replication dispute in print

Avramov et al. (2023): the CPZ signal's FF6 alpha is **71% lower** without microcaps.

CPZ (fn. 45): they "do not consider our SDF portfolio based on ω , but use the SDF loadings β to construct a long-short portfolio ... Thus, they study different portfolios."

CPZ shared their data and estimated models with Avramov et al.

Which portfolio tests the paper's claim? Keep this question for Block D.

Deconstruction prompts

1. What fails if we only minimize forecast MSE under a low SNR?
2. What do adversarial test assets buy intellectually?
3. Is this ML, or "econometrics with NN function classes"?
4. What would a replication need? (CRSP + characteristics, 178 macro series, GPU time, the authors' code)

Homework sketch (notebook 04)

A toy illustration, **not** a replication:

- Start from Gu-style forecasts
- Add a simple penalty that encourages economically coherent long–short behaviour
- Compare pure-MSE ranking with constrained ranking on synthetic data

Put the penalty **inside** training, not after it.

Takeaway

After the ML horse race, the frontier question is **which economic restrictions to encode in the objective.**

Next: Block D, replication and robustness as research questions